



ETHERNET 테스트

CRZ Technology





Mango-M32F2 Ethernet 테스트

Ethernet pinmap

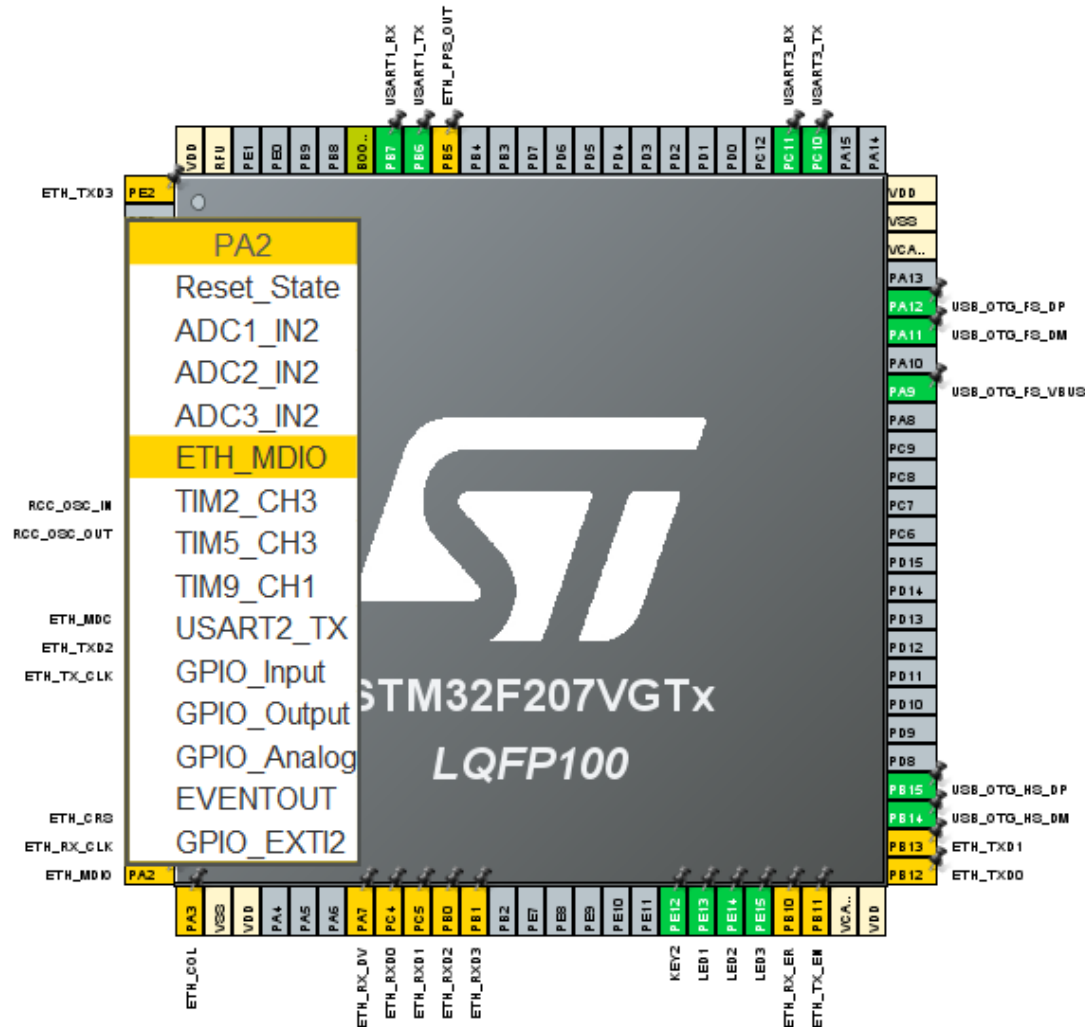
■ Ethernet – PA0~3, PA7, PB0~1, PB5, PB10~13, PC1~5, PE2

Pin Name	Pin Function
PC3	MII Transmit Clock(MII_TX_CLK)
PB11	MII Transmit Enable(MII_TX_EN)
PB12	MII Transmit Data(MII_TXD0)
PB13	MII Transmit Data(MII_TXD1)
PC2	MII Transmit Data(MII_TXD2)
PE2	MII Transmit Data(MII_TXD3)
PA1	MII Receive Clock(MII_RX_CLK)
PA3	MII Collision Detect(MII_COL)
PA0	MII Carrier Sense(MII_CRS)
PA7	MII Receive Data Valid(MII_RX_DV)
PC4	MII Receive Data(MII_RXD0)
PC5	MII Receive Data(MII_RXD1)
PB0	MII Receive Data(MII_RXD2)
PB1	MII Receive Data(MII_RXD3)
PB10	MII Receive Error(MII_RX_ER)
PB5	MII PPS Out(MII_PPS_OUT)
PC1	MII MDC(MII_MDC)
PA2	MII MDIO(MII_MDIO)

Pinout view

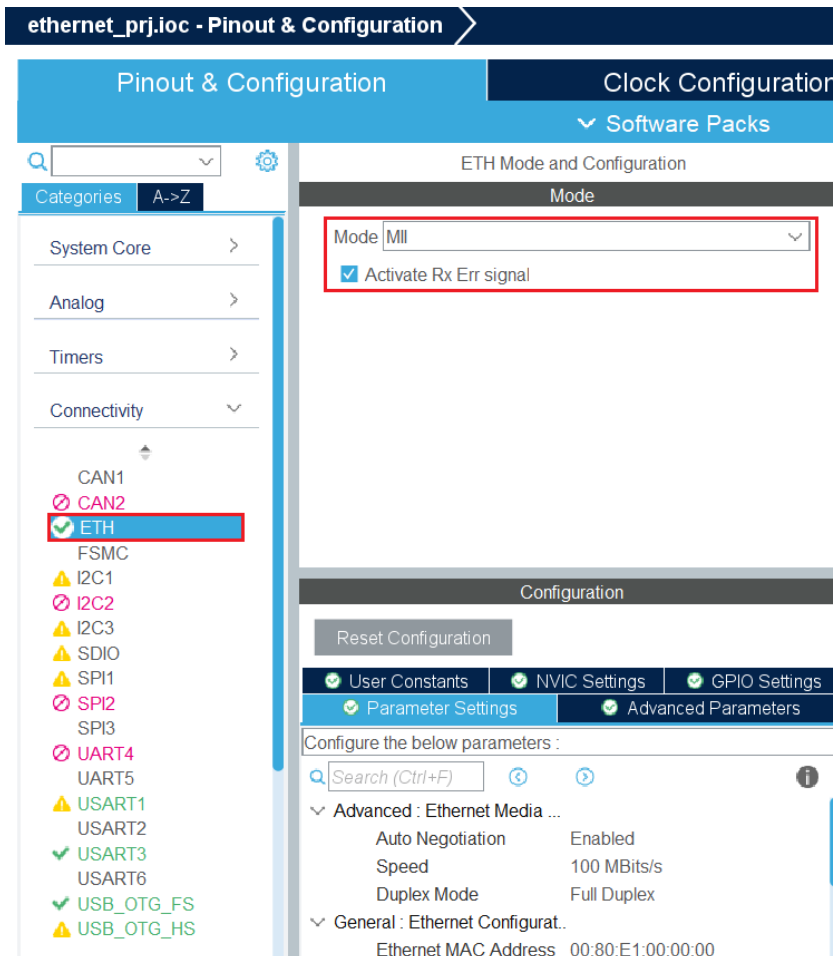
Pinout view

System view



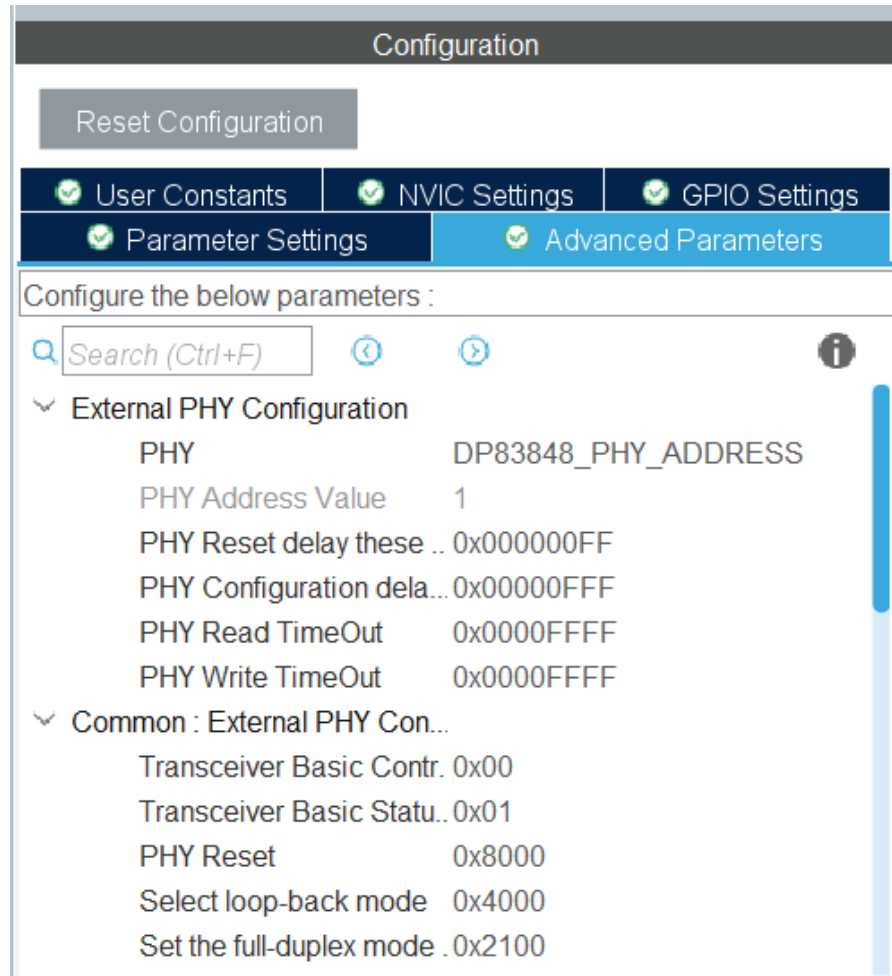
ETH 설정

- “Connectivity” 클릭, “ETH” 클릭.
- “MII”, “Activate Rx Err signal” 클릭.



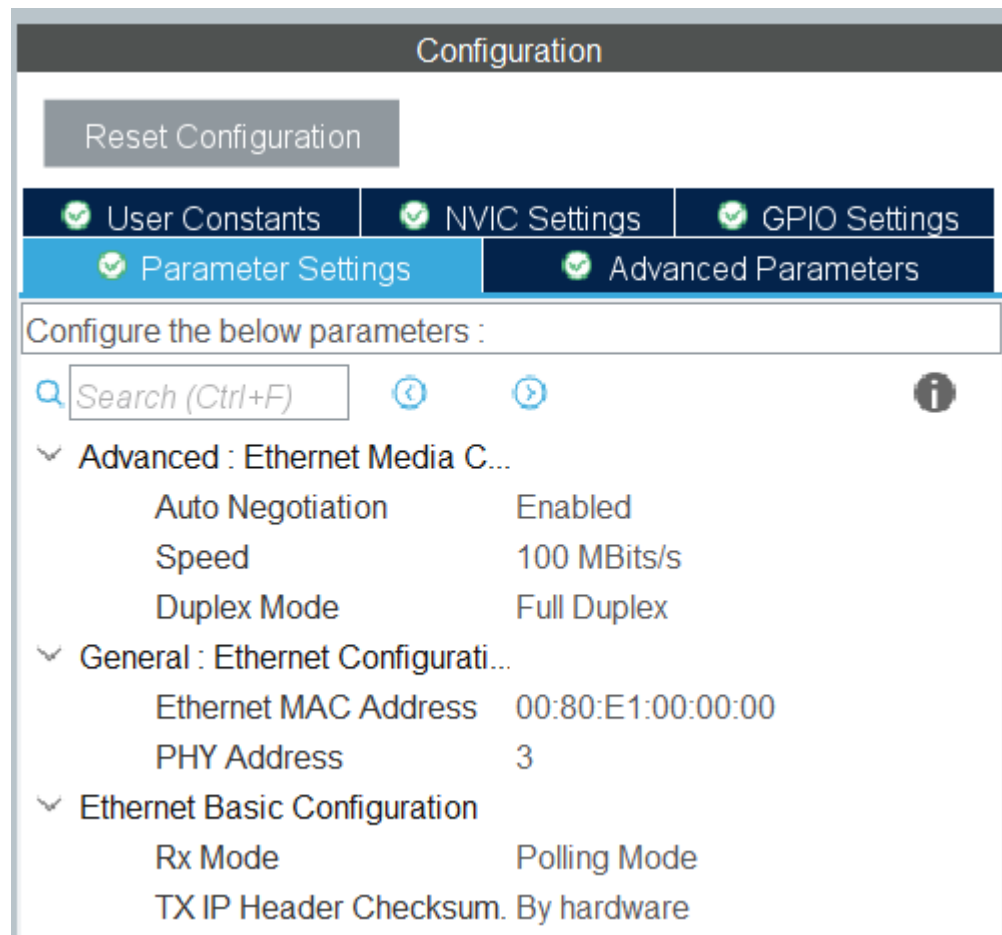
ETH 설정(2)

- “Configuration” -> “Advanced Parameters” 클릭.
- Mango-M32F2_V1.2, RTL8201F 이더넷 PHY인 경우 PHY ADDRESS를 1로 선택.



ETH 설정(3)

- “Configuration” -> “Parameter Settings” 클릭.
- Mango-M32F2_V2.0, KSZ8081MNX 이더넷 PHY인 경우 PHY ADDRESS를 3으로 선택.



LWIP 설정

- “Middleware and Software Packs“ 클릭, “LWIP” 클릭, “Enabled“ 클릭.
- “LWIP_DHCP”-”Disabled”클릭, “IP Address Settings“ 편집.

The screenshot displays the 'ethernet_prj.ioc - Pinout & Configuration' window. The left sidebar shows a tree view of 'Middleware and Software Packs' with 'LWIP' selected and highlighted with a red box. The main panel is titled 'Pinout & Configuration' and 'Clock Configuration'. Under 'Software Packs', the 'LWIP Mode and Configuration' section is visible. The 'Mode' is set to 'Enabled', which is also highlighted with a red box. Below this, the 'Configuration' section shows various options like 'Checksum', 'Debug', 'User Constants', etc. The 'General Settings' tab is active, showing 'LwIP Version' as 2.0.3. Under 'IPv4 - DHCP Options', 'LWIP_DHCP (DHC...' is set to 'Disabled'. Under 'IP Address Settings', the 'IP_ADDRESS (IP A...' is set to '192.168.001.200', 'NETMASK_ADDRE...' is set to '255.255.255.255', and 'GATEWAY_ADDRE...' is set to '192.168.001.001'. These settings are highlighted with a red box.

LWIP 설정

- “Middleware and Software Packs“ 클릭, “LWIP” 클릭.
- “HTTPD“ 클릭, “LWIP HTTPD”-“Enabled” 클릭.

The screenshot displays the 'ethernet_prj.ioc - Pinout & Configuration' window. The left sidebar shows a list of 'Middleware and Software Packs' under the 'Categories' tab, with 'LWIP' selected and highlighted by a red box. The main panel is divided into two sections: 'Pinout & Configuration' and 'Clock Configuration'. The 'Software Packs' section is expanded, showing 'LWIP Mode and Configuration'. Under the 'Mode' tab, 'Enabled' is checked. The 'Configuration' tab is active, showing a grid of options: Checksum, Debug, User Constants, MDNS/TFTP, Perf/Checks, Statistics, IPv6, HTTPD (highlighted with a red box), SNMP, and SNTP. Below this, the 'Configure the below parameters :' section is visible, with 'Show Advanced Parameters' checked. Under 'HTTPD Options', 'LWIP_HTTPD (LwIP HTTP...)' is highlighted with a red box and set to 'Enabled'. Other options like 'LWIP_HTTPD_CGI (HTTP ...)' are set to 'Disabled'.

Ethernet Test Project

- Code 생성 후 main.c 수정 - header추가.
- httpd_init() / MX_LWIP_Process()호출 추가

```
26 /* Private includes -----*/
27 /* USER CODE BEGIN Includes */
28 #include "lwip/apps/httpd.h"
29 /* USER CODE END Includes */

130 MX_LWIP_Init();
131 /* USER CODE BEGIN 2 */
132 printf("Ethernet test\r\n");
133 httpd_init();
134 /* USER CODE END 2 */

135
136 /* Infinite loop */
137 /* USER CODE BEGIN WHILE */
138 while (1)
139 {
140     MX_LWIP_Process();
141     /* USER CODE END WHILE */
142     MX_USB_HOST_Process();
143
144     /* USER CODE BEGIN 3 */
145 }
146 /* USER CODE END 3 */
147 }
```

Ethernet Test Project

■ 이미지 빌드 시 에러 발생.

```
../Middlewares/Third_Party/LwIP/src/apps/httpd/fs.c:43:10: fatal error: fsdata.c: No such file or directory
  43 | #include "fsdata.c"
      |          ^~~~~~
compilation terminated.
make: *** [Middlewares/Third_Party/LwIP/src/apps/httpd/subdir.mk:22: Middlewares/Third_Party/LwIP/src/apps/httpd/fs.o] Error 1
make: *** Waiting for unfinished jobs....
"make -j8 all" terminated with exit code 2. Build might be incomplete.
```

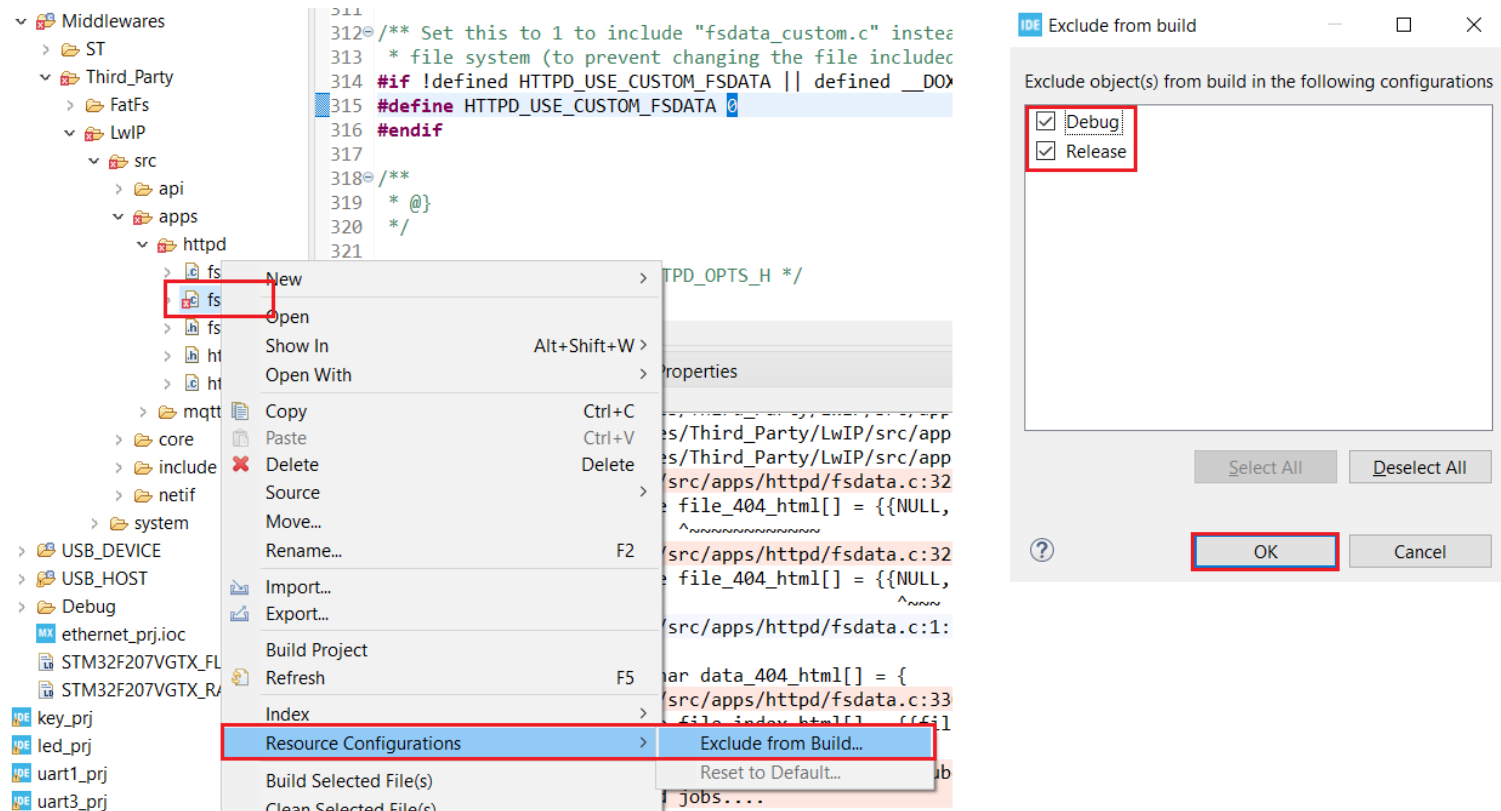
■ fsdata.c를 Middlewares -> Third_Party -> LwIP -> src -> apps -> httpd 폴더에 복사.



fsdata.c

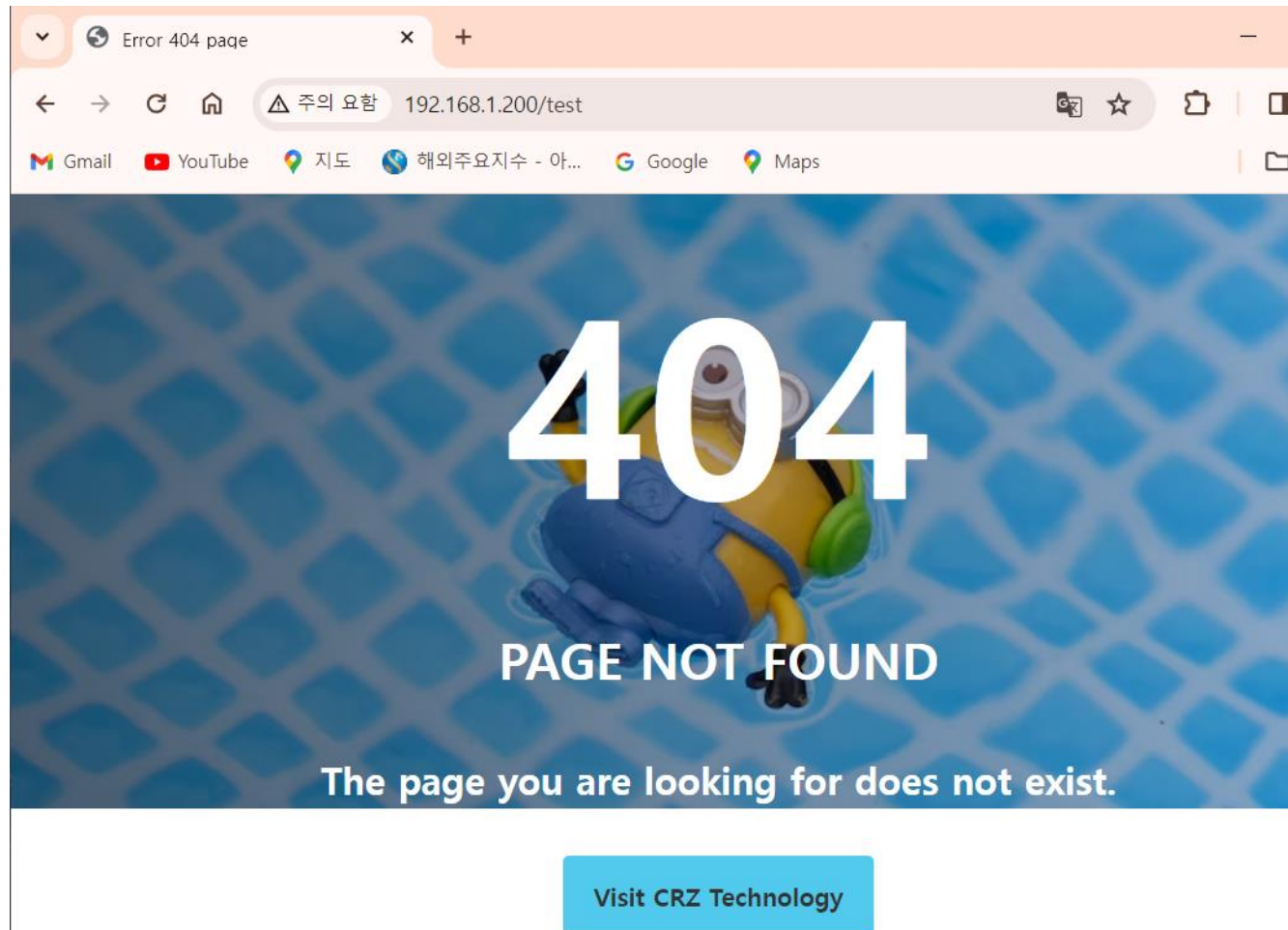
Ethernet Test Project

- fsdata.c를 빌드하지 않도록 설정 - fsdata.c 우클릭 후 "Resource Configurations"->"Exclude from Build..." - "Debug"/"Release" 선택 후 "OK" 클릭.



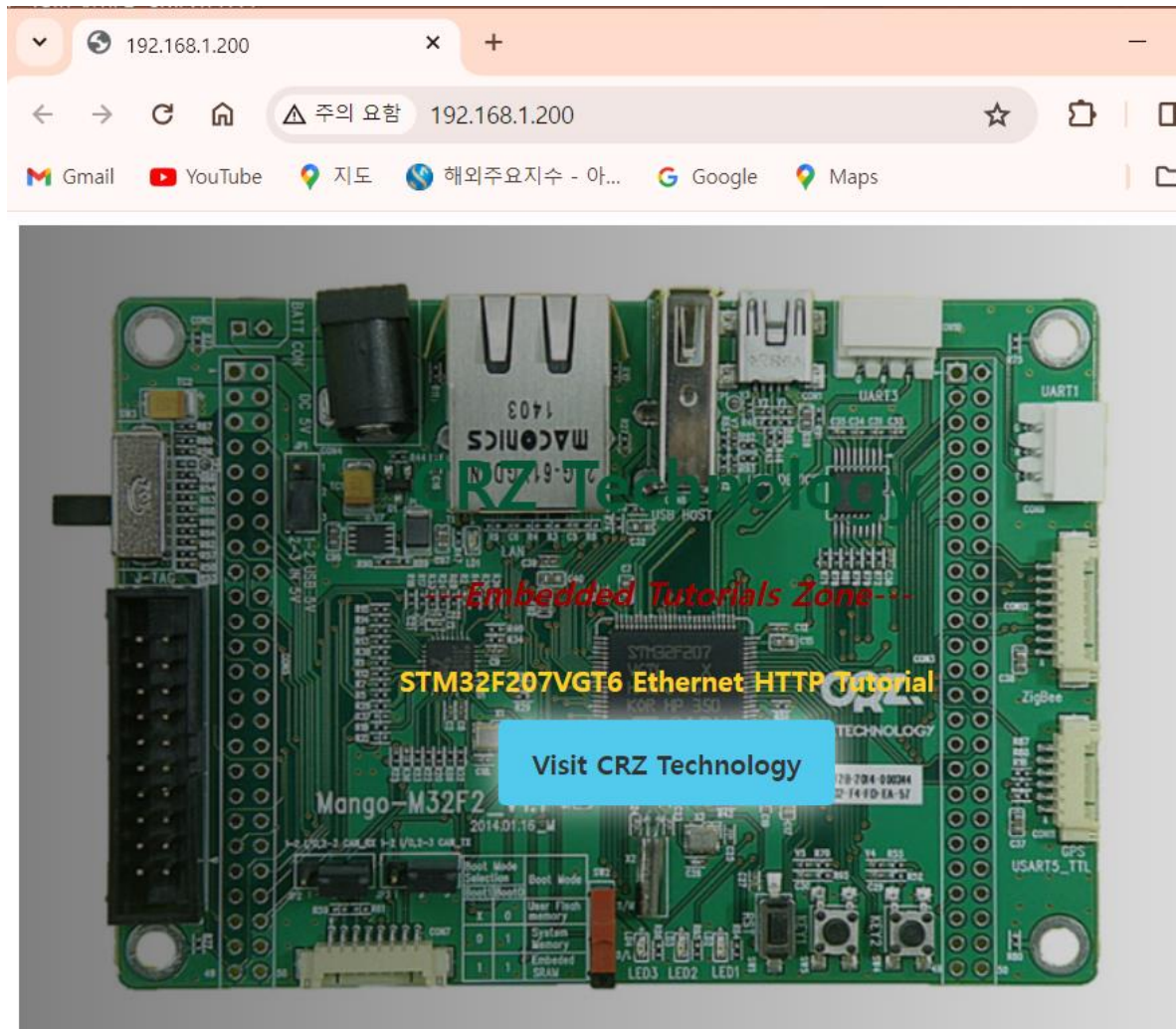
Test 결과 확인

- <http://192.168.1.200/test> 접속시.



Test 결과 확인

- <http://192.168.1.200/> 접속시.



Thank You

